DATA STRUCTURE
LAB PROGRAMS

10. Write a C program to implement Linked list operations

```
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

struct node *head = NULL;

// Insert at Beginning
void insertBeg(int val) {
    struct node *newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = val;
    newnode->next = head;
    head = newnode;
}

// Insert at End
void insertEnd(int val) {
    struct node *newnode = (struct node*)malloc(sizeof(struct node));
```

```
    struct node *temp = head;

    newnode->data = val;
    newnode->next = NULL;

    if(head == NULL) {
        head = newnode;
        return;
    }

    while(temp->next != NULL)
        temp = temp->next;

    temp->next = newnode;
}

// Insert at Any Position
void insertPos(int val, int pos) {
    struct node *newnode = (struct node*)malloc(sizeof(struct node));
    struct node *temp = head;
    int i;

    newnode->data = val;

    for(i = 1; i < pos - 1; i++)
```

```
        temp = temp->next;

    newnode->next = temp->next;
    temp->next = newnode;
}

// Delete from Beginning
void deleteBeg() {
    struct node *temp = head;

    if(head == NULL)
        return;

    head = head->next;
    free(temp);
}

// Delete from End
void deleteEnd() {
    struct node *temp = head, *prev;

    if(head == NULL)
        return;

    if(head->next == NULL) {
```

```
        free(head);
        head = NULL;
        return;
    }

    while(temp->next != NULL) {
        prev = temp;
        temp = temp->next;
    }

    prev->next = NULL;
    free(temp);
}

// Delete from Any Position
void deletePos(int pos) {
    struct node *temp = head, *prev;
    int i;

    for(i = 1; i < pos; i++) {
        prev = temp;
        temp = temp->next;
    }

    prev->next = temp->next;
```

```c
    free(temp);
}

// Display
void display() {
    struct node *temp = head;

    while(temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}

int main() {
    insertBeg(20);
    insertBeg(10);
    insertEnd(40);
    insertPos(30, 3);

    printf("After Insertions:\n");
    display();

    deleteBeg();
    deleteEnd();
```

```
    deletePos(2);

    printf("After Deletions:\n");
    display();

    return 0;
}
```

```
After Insertions:
10 -> 20 -> 30 -> 40 -> NULL
After Deletions:
20 -> NULL


...Program finished with exit code 0
Press ENTER to exit console.
```